

The Hubble Calibration Ratio and the Cosmological Constant

One finite covering ladder, one exact ratio, and one positive vacuum scale

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22 July 2026 · Publication edition 3.0

[doi:10.5281/zenodo.21279104](https://doi.org/10.5281/zenodo.21279104)

EXACT CALIBRATION RESULT

$R = 3305 / 3048 = 1.084317585301837\dots$; approximately five parts per million from the registered observational ratio.

The ratio is derived from the fold's counted shallow and deep ends; measurements enter only after the exact fraction is sealed.

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Contents

One finite covering ladder, one exact ratio, and one positive vacuum scale

Central results

1. The empirical ratio being addressed
2. The shallow end of the covering ladder
 - 2.1 Vacuum and matter are the fold orbit
 - 2.2 The colour-depth tower
3. The deep end and the exact calibration ratio
4. Empirical comparison
5. The cosmological constant on one scale axis
6. Machine-checked evidence
7. Falsification conditions
8. Conclusion

References and provenance

Abstract

The nearby distance ladder and the cosmic-microwave-background inference assign different values to the Hubble constant. Using the registered SH0ES 2022 value 73.04 ± 1.04 km/s/Mpc and the final Planck base-Lambda-CDM value 67.36 ± 0.54 km/s/Mpc, the observed central-value ratio is

$$R_{\text{(observed)}} = 73.04/67.36 = 1.08432304038....$$

Smithian Fold Theory derives a corresponding calibration ratio without importing either measurement. The fold supplies binary count $b=2$ and colour count $c=3$. Their period-two orbit partitions the One into vacuum $2/3$ and matter $1/3$; the colour-depth binary tower is $2^3=8$; and the leading calibration is uniquely

$$R_1 = 1 + (2/3)/(8) = (13)/(12).$$

The same covering ladder has a forced deep end. The minimal binary cover of $3^4=81$ is depth seven, whose genuine period-seven orbit floor is $2^7-1=127$. Reading the identical calibration form at that end gives

$$R_2 = 1 + (2/3 + 1/127)/(8) = (3305)/(3048) = 1.08431758530....$$

The difference from the registered observational ratio is approximately five parts per million. Exact machine checks establish the partition, the tower, the genuine orbit floor, uniqueness of the leading form and uniqueness of the second-order deepening. The measurements enter only after the ratio is derived.

The same foundation also fixes the vacuum as the positive half-One and places it on one counted scale axis with exponent $127/2$. The corpus therefore does not construct the conventional zero-point mode sum whose cancellation produces the cosmological-constant discrepancy. It derives one positive vacuum state on one finite hierarchy.

Central results

Quantity	Exact SFT result	Role
Vacuum share	$2/3$	Period-two upper part
Matter share	$1/3$	Period-two lower part
Flat partition	$2/3 + 1/3 = 1$	Exact partition of the One
Colour-depth tower	$2^3 = 8$	Shallow calibration scale
Leading ratio	$13/12$	Unique admitted leading form
Deep covering depth	7	Minimal binary cover of 3^4
Deep orbit floor	127	2^7-1 , exact fold period seven
Terminal calibration read	$3305/3048$	Unique admitted deepening
Positive vacuum	$1/2$	Unique self-antipodal ground
Scale exponent	$127/2$	One finite hierarchy axis

1. The empirical ratio being addressed

The comparison is between two different observational routes. The SH0ES distance ladder reports a local value

$$H_{(0,\text{late})} = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1},$$

while the final Planck analysis under base-Lambda-CDM reports

$$H_{(0,\text{early})} = 67.36 \pm 0.54 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

This paper registers their central-value ratio as its comparison target:

$$R_{\text{(observed)}}=(H_{\text{(0,late)}})/(H_{\text{(0,early)}})=(73.04)/(67.36)=1.08432304038....$$

The distinction between derivation and comparison is enforced. Neither Hubble value, its uncertainty nor their ratio appears in the functions that construct the SFT result. They occur only in the comparison test and publication evidence.

2. The shallow end of the covering ladder

2.1 Vacuum and matter are the fold orbit

The self-proven theorem forces the One and the fold. The fold's first non-trivial period has binary count two, and the next structural orbit gives colour count three. The period-two orbit is

$$1/3 \rightarrow 2/3 \rightarrow 1/3.$$

The two orbit parts exhaust the One:

$$f_{\text{(matter)}}=1/3, f_{\text{(vacuum)}}=2/3, f_{\text{(matter)}}+f_{\text{(vacuum)}}=1.$$

The construction therefore reads the cosmic budget as a structural partition before any cosmological measurement is consulted.

2.2 The colour-depth tower

The binary generator at colour depth produces

$$T=b^c=2^3=8.$$

The leading calibration distributes the vacuum share over this tower and adds it to the One:

$$R_1=1+(f_{\text{(vacuum)}})/(T) = 1+(2/3)/(8) = (13)/(12).$$

The engine does not merely calculate this fraction. It enumerates the admitted same-size forms over vacuum, matter, tower and One. The forms $1-\text{vacuum}/\text{tower}$, $1+\text{matter}/\text{tower}$, $\text{vacuum}+1/\text{tower}$, $1+\text{vacuum}*\text{tower}$ and $1+\text{matter}*\text{tower}$ do not coincide. Exactly one form reaches $13/12$; the forced_unique guard halts otherwise.

3. The deep end and the exact calibration ratio

The next generational volume is $c^{(c+1)}=3^4=81$. Its minimal binary covering depth is seven:

$$2^6 < 81 \leq 2^7.$$

The same depth is independently obtained from $c+(c+1)=3+4=7$; the engine requires those routes to agree.

The relevant recurrent object is not the pre-periodic tower $2^7=128$, but the orbit floor

$$P=2^7-1=127.$$

The exact fold-period engine verifies that the unit fraction $1/127$ has period seven. The deep-end reading sharpens the same vacuum term by one recurrence over that floor:

$$f_{\text{(vacuum,deep)}}=2/3+(1)/(127).$$

No new outer formula is introduced. The same calibration architecture gives

$$R_2=1+(f_{\text{(vacuum,deep)}})/(8) = 1+(2/3+1/127)/(8) = (3305)/(3048) = 1.084317585301837....$$

The engine tests the possible one-level deepenings $\text{vacuum}+1/\text{floor}$, $\text{vacuum}-1/\text{floor}$, $\text{vacuum}+1/\text{tower}$ and $\text{vacuum}+1/\text{depth}$. Exactly the genuine orbit-floor construction reaches the registered exact fraction.

4. Empirical comparison

Route	Ratio
SH0ES 2022 / Planck 2018 central values	1.084323040380...
SFT leading result	1.083333333333...
SFT deep-end result	1.084317585302...

The absolute difference between the registered observed central-value ratio and the SFT deep-end result is approximately

$$5.45508 \times 10^{-6},$$

or approximately 5.03 parts per million of the observed ratio.

This agreement is not used to choose among constructions. The SFT fraction is fixed by the period counts, minimal cover, orbit floor and uniqueness guards. Observation judges the already-derived result.

The physical interpretation stated by the model is direct: the late and early determinations read the shallow and deep ends of one counted covering hierarchy. The ratio is therefore a scale-transition relation, not a new fitted cosmological parameter.

5. The cosmological constant on one scale axis

The vacuum state is separately forced from the displaced ground:

$$x+x=1 \Rightarrow x=1/2.$$

It is positive, self-antipodal and a preimage of the One:

$$1-1/2=1/2, F(1/2)=1.$$

The corpus's absolute-scale construction counts $2^7-1=127$ massive states and the half-One coupling. Their exact hierarchy exponent is

$$E=127 \times 1/2=(127)/(2).$$

This is one finite scale axis. The derivation never forms an unbounded sum of continuum zero-point modes and therefore never creates a quantity requiring cancellation through 120 decimal orders. Its cosmological statement is a positive vacuum on one counted hierarchy, linked to the same depth-seven structure used in the Hubble calibration.

6. Machine-checked evidence

The readable sources are:

- constants/hubble_tension.ep - partition, tower, both exact ratios, orbit-floor check and both uniqueness guards;
- constants/cosmological_constant.ep - positive half-One vacuum and single-scale-axis identity;
- constants/absolute_scale.ep - the exact $127/2$ hierarchy exponent;
- constants/grand_lock.ep - simultaneous perturbation of the Hubble ratio with the other colour-dependent constants;
- constants/counterfactual_map.ep - classification of necessary, contingent and conventional quantities.

The focused certificates are:

```
./verify/test_hubble_tension
./verify/test_cosmological_constant
./verify/test_grand_lock
./verify/test_counterfactual_map
```

The current-source corpus gate independently completes **326 suites and 2,002 checks with zero failures**, with all 326 generated certificates byte-identical to their committed forms.

7. Falsification conditions

The result fails within its stated construction if:

- the fold does not supply binary two and colour three;
- 1/3 and 2/3 do not form the exact period-two partition;
- the colour-depth tower is not eight;
- another admitted leading form reaches 13/12;
- the minimal binary cover of 81 is not seven;
- 1/127 does not have fold period seven;
- another admitted deepening reaches 3305/3048;
- the generated certificate differs from the readable source;
- a future registered early/late calibration ratio excludes the exact SFT relation under the specified observational pairing.

No fitted coefficient is available to repair such a failure.

8. Conclusion

The Hubble calibration ratio and the positive vacuum scale are two readings of one finite covering architecture. Binary two and colour three partition the One, supply tower eight and force the leading ratio 13/12. The deep covering depth is seven; its genuine orbit floor is 127; and the unique deep-end read is

$$R=3305/3048=1.084317585301837....$$

Against the registered SH0ES-to-Planck central-value ratio, the exact result agrees to approximately five parts per million. The vacuum is independently the positive half-One on the single 127/2 scale axis. Both results are machine-checked, parameter-free and exposed to measurement after derivation.

References and provenance

- Main corpus: <https://github.com/MettaMazza/Smithian-Fold-Theory-Of-Everything>
- Theory concept DOI: <https://doi.org/10.5281/zenodo.21182468>
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Scientific author and publication authority: **Maria Smith**. Publication audit and document engineering assistance: **OpenAI Codex, GPT-5**. Agent assistance does not transfer authorship of scientific claims or authority to define the corpus's forcing and derivation validity.